

GED Science Practice Test 7

PART A — QUESTIONS

GED-SCI7-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student studies pill bug behavior. Four connected chambers contain different moisture levels, but each chamber has the same temperature and amount of light. The student places 40 pill bugs in the center and counts how many are in each chamber after 15 minutes.

Chamber moisture | Pill bugs after 15 min

Very dry | 3

Slightly dry | 7

Moist | 22

Very wet | 8

Which conclusion is best supported by the data?

- A) Pill bugs preferred the moist chamber over the other chambers.
- B) Pill bugs preferred the very dry chamber over the other chambers.
- C) Moisture had no relationship to pill bug location.
- D) All pill bugs moved to the very wet chamber.

GED-SCI7-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student creates waves in a rope and records the wave amplitude and energy transferred.

Amplitude (cm): 1, 2, 3, 4

Energy transferred (J): 2, 8, 18, 32

Which statement is best supported by the data?

- A) Greater amplitude is associated with more energy transferred.
- B) Greater amplitude is associated with less energy transferred.
- C) Energy transferred stays the same for all amplitudes.
- D) Amplitude and energy are unrelated because waves do not carry energy.

GED-SCI7-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A soil profile has several layers. The top layer contains many decayed leaves and roots. Below it is

a darker layer rich in humus. Deeper layers contain weathered rock fragments and less organic material.

Which layer would most likely support the greatest plant growth?

- A) The upper layer rich in decayed organic material
- B) The deepest layer with mostly rock fragments
- C) A layer with no nutrients or water
- D) A layer made only of solid bedrock

GED-SCI7-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Human body cells usually contain 46 chromosomes. During meiosis, cells divide to form sperm or egg cells. These reproductive cells contain half the normal chromosome number, so the normal number is restored when sperm and egg join.

How many chromosomes are usually found in a human sperm or egg cell?

- A) 23
- B) 46
- C) 69
- D) 92

GED-SCI7-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

Momentum can be calculated as $\text{mass} \times \text{velocity}$.

A rolling cart has a mass of 8 kg and a velocity of 3 m/s.

What is the cart's momentum in $\text{kg} \cdot \text{m/s}$?

Type your answer as a whole number.

GED-SCI7-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how ramp surface affects the time needed for a toy car to roll down a ramp. The student tests cardboard, felt, and smooth plastic surfaces. The ramp angle and starting position are kept the same.

Which TWO factors are controlled variables?

- A) Ramp angle
- B) Starting position of the toy car
- C) Ramp surface
- D) Time needed to roll down
- E) Final measured result

GED-SCI7-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In humans, biological sex is usually affected by sex chromosomes. A typical egg cell carries an X chromosome. A sperm cell may carry either an X or a Y chromosome. If an egg with X joins with a sperm carrying Y, the offspring usually has [D1] chromosomes and is genetically [D2].

Dropdowns:

D1) XY | XX | YY

D2) male | female | identical to the mother

GED-SCI7-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A metal lid on a glass jar becomes easier to loosen after warm water is run over the lid. Heating the metal causes its particles to move more and spread slightly farther apart, so the lid expands.

Which process best explains why the lid becomes easier to remove?

- A) Thermal expansion
- B) Nuclear decay
- C) Photosynthesis
- D) Condensation inside the metal

GED-SCI7-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

An aquifer is an underground layer that can store and transmit groundwater. A community builds a well in a sandy aquifer because water can move through connected pore spaces in the sand. A nearby layer of solid clay allows much less water to pass through.

Which property makes the sandy aquifer useful for a well?

- A) High permeability
- B) High fossil content
- C) Lack of gravity
- D) Absence of pore spaces

GED-SCI7-010

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Blood calcium levels are partly regulated by hormones. When blood calcium becomes too low, parathyroid hormone helps increase calcium in the blood. When calcium returns to a normal range, less parathyroid hormone is released.

Which idea is best illustrated by this process?

- A) Negative feedback helps maintain homeostasis.
- B) Hormones prevent cells from responding to changes.
- C) Calcium levels can never change in the body.
- D) The digestive system controls all nerve impulses.

GED-SCI7-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Number of wire coils around the nail
- B) Number of paper clips picked up by the electromagnet
- C) Same battery used in each trial
- D) Color of the paper clips

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI7-012

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student grows radish plants alone and with ryegrass plants to study competition. Each pot receives the same soil, water, and light. After three weeks, the student measures the average radish mass.

Growth condition | Average radish mass (g)

Radish alone | 12.4

Radish with ryegrass | 7.1

Which conclusion is best supported by the data?

- A) Ryegrass reduced radish growth, likely by competing for resources.
- B) Ryegrass increased radish growth by producing extra sunlight.

- C) Radish plants grew the same mass in both conditions.
- D) Competition cannot occur between plants.

GED-SCI7-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student places a dot of black marker ink near the bottom of a strip of paper. The paper is placed in a small amount of water, but the ink dot stays above the water level. As water moves up the paper, different colors in the ink separate because they move at different speeds.

Which method is being used?

- A) Paper chromatography
- B) Freezing
- C) Magnetism
- D) Electrical grounding

GED-SCI7-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A group of 16 solar panels produces 64 kilowatt-hours of electrical energy on a sunny day.

What is the average electrical energy produced per panel?

Type your answer as a whole number.

GED-SCI7-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A sports scientist compares heart-rate recovery after exercise. Two groups complete the same 5-minute stepping activity. Group A exercises regularly, and Group B does not. Heart rate is measured one minute after exercise ends.

Group | Average heart rate 1 min after exercise (beats/min)

- A | 96
- B | 122

Which claim is best supported by the data?

- A) Group A recovered to a lower heart rate one minute after the same exercise.
- B) Group B had a lower heart rate than Group A after exercise.
- C) Exercise regularity had no possible relationship to recovery heart rate.
- D) Both groups had exactly the same recovery heart rate.

GED-SCI7-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student investigates how parachute size affects falling time. The student attaches identical masses to parachutes with different surface areas and drops each parachute from the same height.

Why should the drop height be kept the same?

- A) To make parachute size the main variable being tested
- B) To make gravity stop acting on the masses
- C) To make every parachute have the same surface area
- D) To prevent air resistance from occurring

GED-SCI7-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A clear window lets most visible light pass through it, so it is [D1]. A wooden door blocks visible light, so it is [D2].

Dropdowns:

D1) transparent | opaque | magnetic

D2) opaque | transparent | dissolved

GED-SCI7-018

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Cells normally divide in a controlled way. When mutations disrupt genes that regulate the cell cycle, some cells may divide too often and form a tumor.

Which statement best explains how tumors can form?

- A) Loss of normal cell-cycle control can lead to uncontrolled cell division.
- B) Cells form tumors only when they stop containing DNA.
- C) All cell division is harmful to living organisms.
- D) Mutations always improve cell-cycle control.

GED-SCI7-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student pulls carts of different masses using the same constant force and records acceleration.

Cart mass (kg): 1, 2, 4, 8
Acceleration (m/s²): 12, 6, 3, 1.5

Which statement is best supported by the data?
A) As mass increased, acceleration decreased.
B) As mass increased, acceleration increased.
C) Acceleration stayed the same for every mass.
D) Mass had no relationship to acceleration.

GED-SCI7-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

Scientists monitor reef water temperature and coral bleaching.

Year	Average summer reef temperature (°C)	Reef area with bleaching (%)
1	28.1	6
2	28.6	12
3	29.4	31
4	29.8	46

Which TWO statements are best supported by the data?
A) Reef bleaching increased as average summer reef temperature increased.
B) The highest bleaching percentage occurred in Year 4.
C) Bleaching decreased every year as temperature increased.
D) Year 1 had the highest reef temperature.
E) Temperature stayed exactly the same in all four years.

GED-SCI7-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Viruses are much smaller than most cells and cannot reproduce on their own. A virus must enter a host cell and use the host cell's machinery to make more virus particles.

Which statement best describes viruses?
A) Viruses require host cells to reproduce.
B) Viruses are made of many organs.
C) Viruses carry out photosynthesis in chloroplasts.
D) Viruses reproduce only by forming seeds.

GED-SCI7-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A bottle contains oil and vinegar. After the bottle sits undisturbed, the liquids separate into layers because they do not mix evenly. Shaking the bottle temporarily spreads tiny droplets of one liquid through the other.

Which statement best describes this mixture?

- A) It is a heterogeneous mixture because the substances are not evenly mixed.
- B) It is a pure element because it contains only one type of atom.
- C) It is a compound because the liquids permanently bond into one substance.
- D) It is a gas because all mixtures become gases.

GED-SCI7-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

Planets closer to the Sun generally complete one orbit in less time than planets farther away. Mercury is closest to the Sun and completes an orbit in about 88 Earth days. Neptune is much farther from the Sun and takes about 165 Earth years to complete one orbit.

Which statement is best supported by this information?

- A) A planet farther from the Sun usually has a longer orbital period.
- B) All planets complete one orbit in exactly 88 Earth days.
- C) Neptune is closer to the Sun than Mercury.
- D) Distance from the Sun has no relationship to orbital period.

GED-SCI7-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A biologist tags 40 turtles in a pond. Later, the biologist captures 50 turtles and finds that 10 of them are tagged.

Estimated population = (number tagged first $\times$ number captured later) $\div$ number tagged in later sample

What is the estimated turtle population?

Type your answer as a whole number.

GED-SCI7-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A circuit has two identical bulbs connected in parallel to a battery. One bulb is unscrewed, but the other bulb stays lit because it still has a complete path connected to the battery.

Which statement best explains this observation?

- A) In a parallel circuit, each branch can provide a separate path for current.
- B) In a parallel circuit, current can flow without a battery.
- C) Removing one bulb turns every wire into an insulator.
- D) Bulbs in parallel cannot use electrical energy.

GED-SCI7-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some cell organelles contain enzymes that break down worn-out cell parts, food particles, or invading bacteria. These organelles help recycle materials inside the cell.

Which organelle is mainly described?

- A) Lysosome
- B) Nucleus
- C) Chloroplast
- D) Cell wall

GED-SCI7-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student claims that increasing the number of coils in an electromagnet makes it stronger. The student wraps wire around a nail 10, 20, and 30 times, connects the same battery, and counts how many paper clips the electromagnet picks up.

Which result would best support the student's claim?

- A) The electromagnet with 30 coils picks up the most paper clips.
- B) The electromagnet with 10 coils picks up the most paper clips.
- C) Each electromagnet picks up exactly the same number of paper clips.
- D) The battery is changed for each number of coils.

GED-SCI7-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A mineral sample scratches gypsum but does not scratch quartz. The Mohs hardness scale lists gypsum as 2 and quartz as 7. A mineral can scratch another mineral only if it is harder.

Which conclusion is best supported?

- A) The mineral sample has a hardness greater than 2 but less than 7.
- B) The mineral sample must have a hardness of exactly 10.
- C) The mineral sample is softer than gypsum.
- D) The mineral sample and quartz have the same hardness.

GED-SCI7-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic compares white blood cell counts before and during a bacterial infection.

Condition | White blood cells per microliter of blood

Before infection | 6,500

During infection | 13,200

Which statement is best supported by the data?

- A) White blood cell count increased during the infection.
- B) White blood cell count decreased to zero during the infection.
- C) White blood cells were absent before the infection.
- D) Infection had no relationship to white blood cell count.

GED-SCI7-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A radioactive medical tracer sample decreases over time.

Time (hours): 0, 6, 12, 18

Amount remaining (mg): 80, 40, 20, 10

What is the half-life of the tracer?

- A) 6 hours
- B) 12 hours
- C) 18 hours
- D) 80 hours

GED-SCI7-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student studies whether repeated gripping affects muscle fatigue. Participants squeeze the same handgrip device 20 times, rest, and then repeat the task. The student records the number of squeezes completed before grip strength drops.

Which TWO steps would improve the investigation?

- A) Use the same handgrip device for all participants.
- B) Give each participant the same rest time between trials.
- C) Let each participant choose a different number of starting squeezes.

- D) Record only the strongest participant's result.
- E) Change the grip device during the study.

GED-SCI7-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records how many vibrations a tuning fork makes in different amounts of time.

Time (s): 1, 2, 3, 4

Vibrations: 30, 60, 90, 120

What is the frequency of the tuning fork?

- A) 15 Hz
- B) 30 Hz
- C) 60 Hz
- D) 120 Hz

GED-SCI7-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A trait is caused by a recessive allele on the X chromosome. In many families, males show the trait more often than females because males usually have only one X chromosome.

Which statement best explains why the trait appears more often in males?

- A) A male with the recessive allele on his only X chromosome will show the trait.
- B) Males always have two X chromosomes that hide recessive alleles.
- C) Females cannot inherit X chromosomes.
- D) X-linked traits are never passed from parents to offspring.

GED-SCI7-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Cells lining parts of the respiratory tract have tiny hairlike structures called cilia. Cilia help move mucus, trapped dust, and microbes away from the lungs.

Which function is best described?

- A) Helping protect the respiratory system from particles and microbes
- B) Producing eggs and sperm
- C) Breaking down food in the stomach
- D) Carrying inherited instructions in chromosomes

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI7-001 — Correct: A — Correct Answer: Pill bugs preferred the moist chamber over the other chambers. — Explanation: The largest number of pill bugs, 22, was found in the moist chamber after 15 minutes.

GED-SCI7-002 — Correct: A — Correct Answer: Greater amplitude is associated with more energy transferred. — Explanation: The data show energy transferred increasing as amplitude increases.

GED-SCI7-003 — Correct: A — Correct Answer: The upper layer rich in decayed organic material — Explanation: Decayed organic material adds nutrients and helps soil support plant growth.

GED-SCI7-004 — Correct: A — Correct Answer: 23 — Explanation: Human reproductive cells usually contain half the chromosome number of body cells, so they contain 23 chromosomes.

GED-SCI7-005 — Correct: 24 — Tolerance: exact — Correct Answer: 24 — Explanation: Momentum = mass $\times$ velocity = 8 kg $\times$ 3 m/s = 24 kg·m/s.

GED-SCI7-006 — Correct: A,B — Correct Answer: Ramp angle | Starting position of the toy car — Explanation: These factors are kept the same so ramp surface is the main variable being tested.

GED-SCI7-007 — Correct: D1=XY; D2=male — Correct Answer: D1=XY; D2=male — Explanation: An X chromosome from the egg and a Y chromosome from the sperm usually produce XY chromosomes and a genetically male offspring.

GED-SCI7-008 — Correct: A — Correct Answer: Thermal expansion — Explanation: Heating causes the metal lid to expand slightly, making it easier to loosen.

GED-SCI7-009 — Correct: A — Correct Answer: High permeability — Explanation: High permeability allows groundwater to move through connected pore spaces toward the well.

GED-SCI7-010 — Correct: A — Correct Answer: Negative feedback helps maintain homeostasis. — Explanation: The hormone response decreases once calcium returns toward normal, which is negative feedback.

GED-SCI7-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Number of wire coils around the nail; T2=Number of paper clips picked up by the electromagnet; T3=Same battery used in each trial — Explanation: The number of coils is changed, the number of paper clips is measured, and the battery is controlled.

GED-SCI7-012 — Correct: A — Correct Answer: Ryegrass reduced radish growth, likely by competing for resources. — Explanation: Radish plants had lower average mass when grown with ryegrass under the same conditions.

GED-SCI7-013 — Correct: A — Correct Answer: Paper chromatography — Explanation: Paper chromatography separates substances in a mixture based on how they move with a solvent.

GED-SCI7-014 — Correct: 4 — Tolerance: exact — Correct Answer: 4 — Explanation: 64 kilowatt-hours $\div$ 16 panels = 4 kilowatt-hours per panel.

GED-SCI7-015 — Correct: A — Correct Answer: Group A recovered to a lower heart rate one minute after the same exercise. — Explanation: Group A had an average heart rate of 96 beats/min, lower than Group B's 122 beats/min.

GED-SCI7-016 — Correct: A — Correct Answer: To make parachute size the main variable being tested — Explanation: Keeping drop height constant helps isolate the effect of parachute size on falling time.

GED-SCI7-017 — Correct: D1=transparent; D2=opaque — Correct Answer: D1=transparent; D2=opaque — Explanation: Transparent materials let light pass through, while opaque materials block visible light.

GED-SCI7-018 — Correct: A — Correct Answer: Loss of normal cell-cycle control can lead to uncontrolled cell division. — Explanation: Tumors can form when mutations disrupt normal regulation of cell division.

GED-SCI7-019 — Correct: A — Correct Answer: As mass increased, acceleration decreased. — Explanation: With the same force, acceleration decreased from 12 m/s^2 to 1.5 m/s^2 as mass increased.

GED-SCI7-020 — Correct: A,B — Correct Answer: Reef bleaching increased as average summer reef temperature increased. | The highest bleaching percentage occurred in Year 4. — Explanation: Bleaching rose from 6% to 46% as temperature increased, and Year 4 had the highest bleaching percentage.

GED-SCI7-021 — Correct: A — Correct Answer: Viruses require host cells to reproduce. — Explanation: Viruses cannot reproduce independently and must use host-cell machinery.

GED-SCI7-022 — Correct: A — Correct Answer: It is a heterogeneous mixture because the substances are not evenly mixed. — Explanation: Oil and vinegar form uneven layers, which is characteristic of a heterogeneous mixture.

GED-SCI7-023 — Correct: A — Correct Answer: A planet farther from the Sun usually has a longer orbital period. — Explanation: The information compares Mercury's short orbit with Neptune's much longer orbit farther from the Sun.

GED-SCI7-024 — Correct: 200 — Tolerance: exact — Correct Answer: 200 — Explanation: Estimated population = $40 \times 50 \div 10 = 200$ turtles.

GED-SCI7-025 — Correct: A — Correct Answer: In a parallel circuit, each branch can provide a separate path for current. — Explanation: Because the bulbs are on separate branches, one bulb can remain lit when the other is removed.

GED-SCI7-026 — Correct: A — Correct Answer: Lysosome — Explanation: Lysosomes contain enzymes that break down and recycle materials inside cells.

GED-SCI7-027 — Correct: A — Correct Answer: The electromagnet with 30 coils picks up the most paper clips. — Explanation: If more coils make the electromagnet stronger, the 30-coil electromagnet should pick up the most paper clips.

GED-SCI7-028 — Correct: A — Correct Answer: The mineral sample has a hardness greater than 2 but less than 7. — Explanation: The sample can scratch gypsum, so it is harder than 2, but it cannot scratch quartz, so it is softer than 7.

GED-SCI7-029 — Correct: A — Correct Answer: White blood cell count increased during the infection. — Explanation: The white blood cell count rose from 6,500 to 13,200 per microliter during infection.

GED-SCI7-030 — Correct: A — Correct Answer: 6 hours — Explanation: The amount decreases by half from 80 mg to 40 mg in the first 6 hours, so the half-life is 6 hours.

GED-SCI7-031 — Correct: A,B — Correct Answer: Use the same handgrip device for all participants. | Give each participant the same rest time between trials. — Explanation: Keeping the device and rest time constant improves fairness and reliability.

GED-SCI7-032 — Correct: B — Correct Answer: 30 Hz — Explanation: Frequency is vibrations per second. The table shows 30 vibrations each second.

GED-SCI7-033 — Correct: A — Correct Answer: A male with the recessive allele on his only X chromosome will show the trait. — Explanation: Because males usually have one X chromosome, a recessive allele on that X chromosome can be expressed.

GED-SCI7-034 — Correct: A — Correct Answer: Helping protect the respiratory system from particles and microbes — Explanation: Cilia move mucus and trapped particles away from the lungs, helping protect the respiratory tract.